Cover letter for npj Genomic Medicine

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Editors

npj Genomic Medicine

Dear Editors,

We submit our Article, **Input matched multisignal colocalization clarifies regulatory assignments at primary biliary cholangitis loci**, for consideration in *npj Genomic Medicine*.

The manuscript addresses a practical problem in genomic medicine: cell-specific regulatory assignments at disease loci can change when single-causal and multi-signal colocalization analyses use different variant support, disease statistics and linkage-disequilibrium inputs. We evaluated a PBC-wide registry of 6,923 locus-gene-cell comparisons, performed source-matched multi-signal analysis in a prespecified 642-comparison high-information subset, and used matched-input analysis to separate two identifiable input effects from residual inferential reclassification.

The main result is bidirectional. After matching support and disease statistics, eight single-causal H4 comparisons supported distinct signal pairs and ten H3 comparisons supported shared signal pairs under the multi-signal analysis. Simulations showed a corresponding trade-off: false-H4 decisions decreased in distinct-signal settings, while shared-signal recovery also decreased. The work therefore presents a reliability and attribution framework rather than claiming general superiority of one method.

The medical-genomics relevance is illustrated in PBC. IL12RB2-NK sharing was reproduced in TenK10K, supported in a different FinnGen disease study system and connected to a disease-linked NK-cell chromatin peak. We explicitly limit this evidence to robust regulatory association: public metadata do not prove person-level independence, the FinnGen analysis is single-causal, direct eQTL-caQTL sharing was not estimated and no perturbational experiment establishes mediation. FCRL3 provides both reproducible B-cell sharing and a covariate-model-sensitive T-cell counterexample. Donor-level liver data support lineage detectability without corrected disease enrichment.

The study fits the journal's scope in integrative disease genomics, immunogenomics and computational analysis of genomic variation. Its contribution is an auditable strategy for deciding when locus-gene-cell assignments remain reliable after input matching, source-aware LD construction and multi-signal decomposition.

All authors approved the manuscript and its submission. The work is original and is not under review elsewhere. The authors declare no competing interests and no specific funding. Ethics approval was not required because the study reanalysed public, de-identified data. Code, frozen protocols and aggregate outputs are publicly available on GitHub. An author-approved open research version has been released publicly; it has not been peer reviewed and does not constitute concurrent journal submission.

OpenAI Codex, operating with the locally installed QiTeng Academic Writing Skill under author supervision, assisted with language editing, document generation and consistency checks. It was not treated as an author or source of scientific evidence. The authors take responsibility for all analyses, interpretations and submitted text.

We have not discussed this manuscript with an editor at *npj Genomic Medicine* before submission.

Sincerely,

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